EMS611U: Conceptual Design of Aircraft

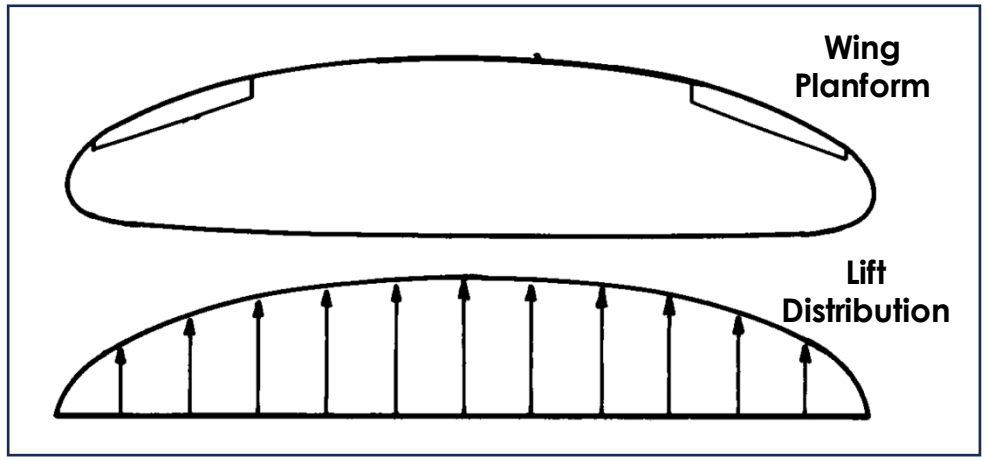
Dr. Hülya Biler
Queen Mary University of London

Week 6

Wing Theory

Twist

- Wing twist is used to prevent tip stall and to revise the lift distribution to approximate an ellipse.
- Typically, wings are twisted between zero and five degrees.



Reminder

It is known that an elliptic wing has the lowest induced drag.

We want our lift distribution to resemble this!

Twist

- Geometric twist
- Aerodynamic twist

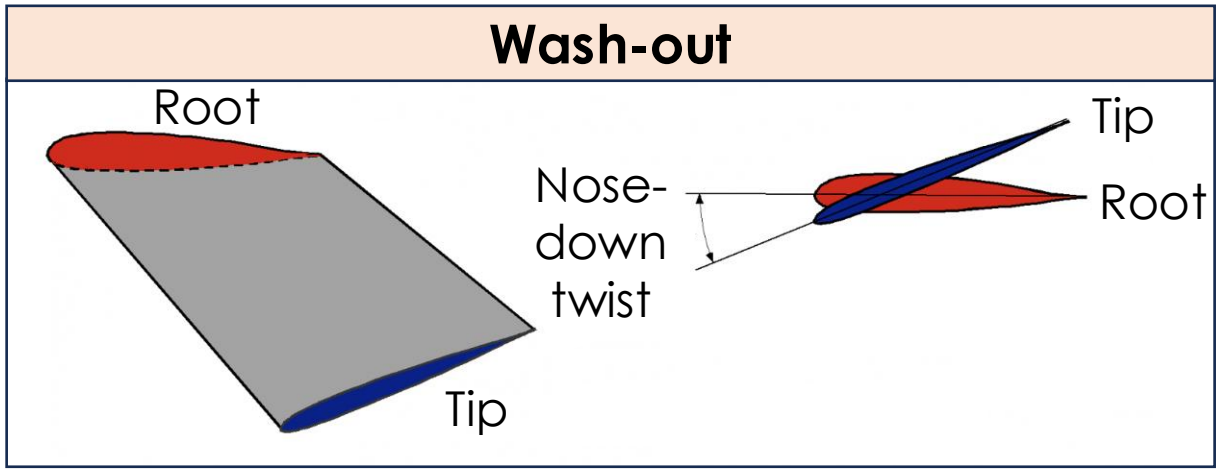
Twist

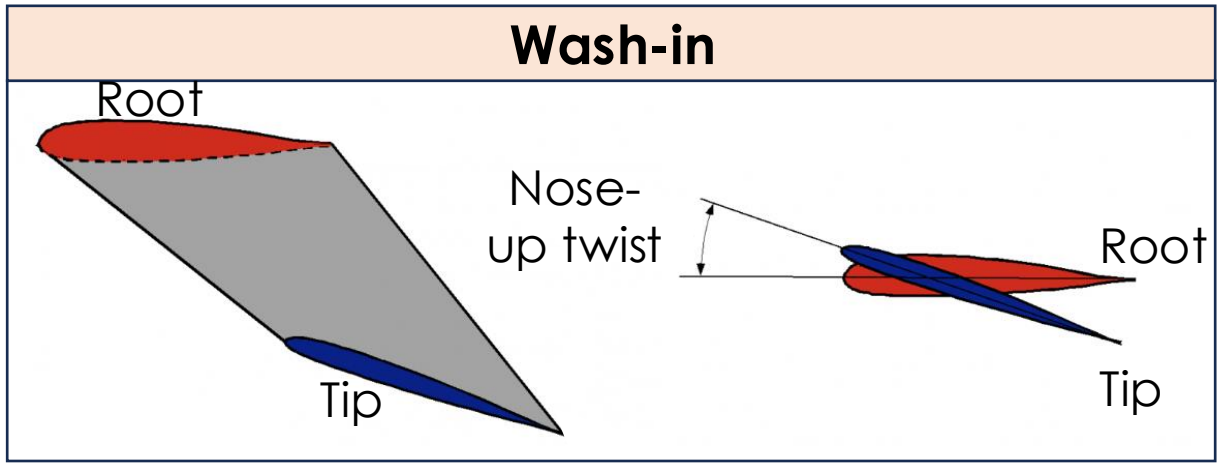
Geometric Twist

The actual change in airfoil angle of incidence, usually measured with respect to the root airfoil.

- If a wing has "linear twist," the twist angle changes in proportion to the distance from the root airfoil.

- A wing whose tip airfoil is at a negative (nose-down) angle compared to the root airfoil is said to have "**wash-out**."
- Although very unusual for an airplane, it is called a "**wash-in**" if a wing is twisted nose-up by increasing its twist from the root to the tip along its span.





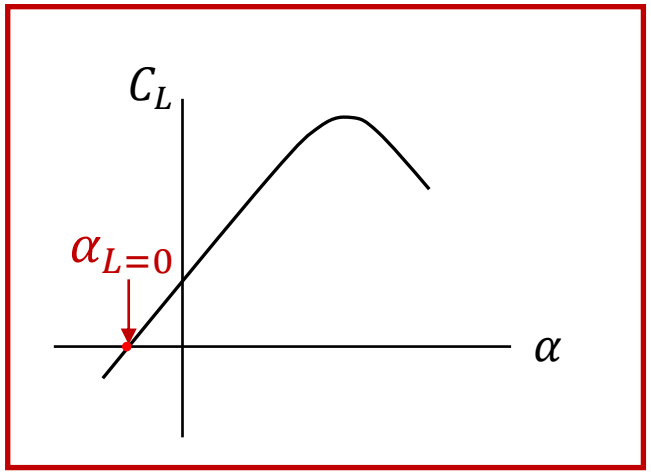
Twist

Aerodynamic Twist

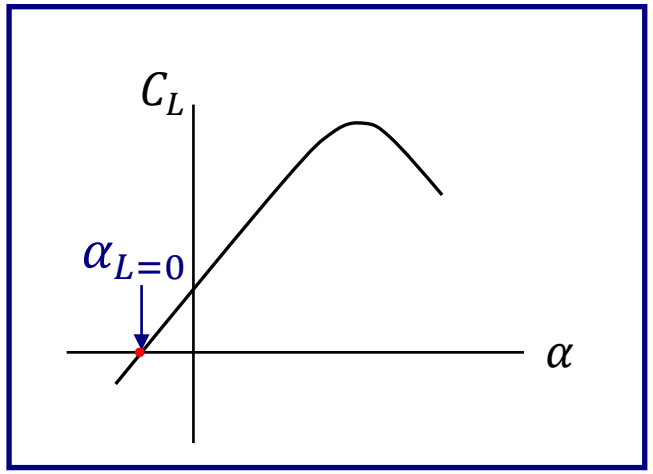
The angle between the zero-lift angle of an airfoil and the zero-lift angle of the root airfoil.

- If the identical airfoil is used from root to tip, the aerodynamic twist is the same as the geometric twist.

Root Aerofoil



Tip Aerofoil



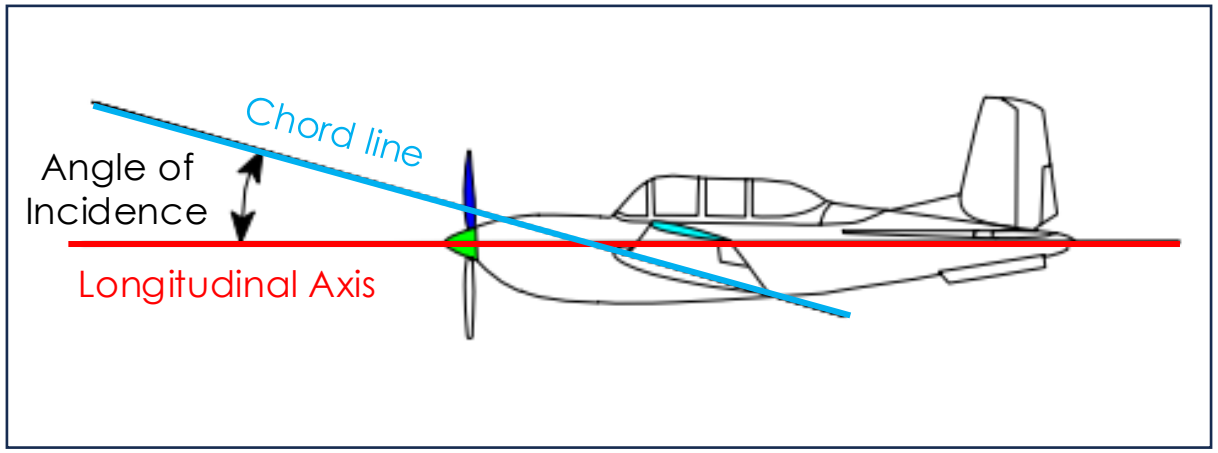
- On the other hand, a wing with no geometric twist can have aerodynamic twist if, for example, the root airfoil is symmetric (zero-lift angle is zero) but the tip airfoil is highly cambered (zero-lift angle is nonzero).

Twist

- When wing twist is used to reshape the lift distribution, the change in lift at some chord station along the span is proportional to the ratio between the new airfoil angle of attack and the original one. Thus, the effect on lift distribution depends upon the original angle of attack of the wing, which in turn depends upon the lift coefficient at which the wing is flying.
- In other words, any attempt to optimize the lift distribution by twisting the wing will be valid only at one lift coefficient. At other lift coefficients, the twisted wing will not get the whole benefit of the twist optimization. The more twist required to produce a good lift distribution at the design lift coefficient, the worse the wing will perform at other lift coefficients. It is for this reason that large amounts of twist (much over 5 deg) should be avoided.
- It is very difficult to optimize twist for an arbitrary wing planform. A computerized solution is employed at large companies. Typically, 3 degrees of twist provides adequate stall characteristics.

Wing Incidence

- The wing incidence angle is the pitch angle of the wing with respect to the fuselage.



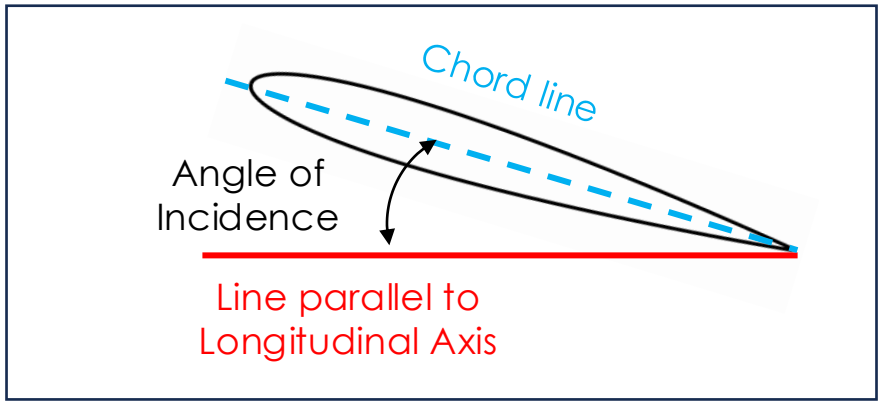
- If the wing is untwisted, the incidence is simply the angle between the fuselage axis and the wing's airfoil chordlines.
- If the wing is twisted, the incidence is defined with respect to some arbitrarily chosen spanwise location of the wing, usually the root of the exposed wing where it intersects the fuselage. Frequently the incidence is given at the root and tip, which then defines the twist as the difference between the two.

Wing Incidence

Why do we need to introduce a wing incidence angle?

To minimize drag at some operating condition, usually cruise.

- The incidence angle is chosen such that when the wing is at the correct angle of attack for the selected design condition, the fuselage is at the angle of attack for minimum drag.
- For a typical, circular straight fuselage, this is approximately zero degrees angle of attack.



What about a passenger aircraft?

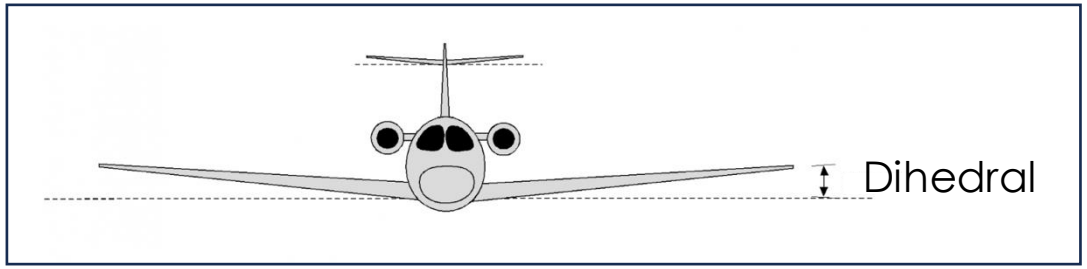
The incidence angle must be carefully chosen to ensure that the flight attendants do not have to push the food carts uphill!

How do we decide on the angle?

- Wing incidence angle is ultimately set using wind tunnel data. For most initial design work, it can be assumed that general aviation and homebuilt aircraft will have an incidence of about 2 degrees, transport aircraft about 1 degree, and military aircraft approximately zero.
- Later in the design process, aero dynamic calculations can be used to check the actual wing incidence angle required during the design condition.

Dihedral

Dihedral angle is the angle that the wings form with respect to the horizon when viewed from the front.



Dihedral helps to maintain aircraft "roll stability"

- A positive dihedral, wingtips angled up, tends to bring an aircraft back to level when it is banked. The counter rolling moment is caused by a sideslip that results from the banking of the aircraft.

For preliminary design purposes:

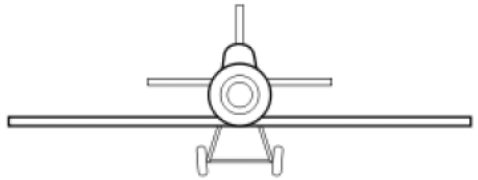
Airplane Type	Dihedral (deg)		
	Wing Location		
	Low	Mid	High
Unswept (Civil)	5 to 7	2 to 4	0 to 2
Subsonic Swept	3 to 7	-2 to 2	-5 to -2
Supersonic Swept	0 to 5	-5 to 0	-5 to 0

- Excessive dihedral can lead to an oscillatory disturbance in motion known as a Dutch roll. Dutch roll is a repeated side to side oscillation that is a result of both yawing and rolling. Such a phenomenon can be disastrous for a craft, but can be countered by increasing vertical tail area.
- This, in turn, will result in an increase in both the weight and drag of the aircraft. Therefore, tail size and dihedral must be considered together to achieve the optimal design for stability, weight, and drag concerns.

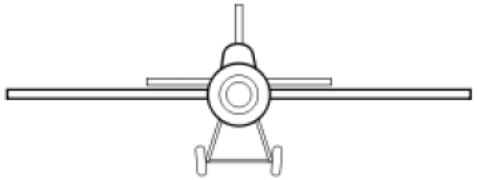
Wing Vertical Location

- There are three choices for the location of the wing on the fuselage namely:

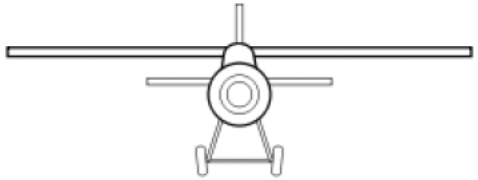
Low-wing Configurations



Mid-wing Configurations



High-wing Configurations



Wing Vertical Location



Advantages:

- Landing gear can be located in the wing thereby avoiding pods on the fuselage and hence lower drag. However to provide adequate ground clearance, the fuselage has to be at a higher level as compared to the high wing configuration.
- Wing structure can be through the fuselage.

Disadvantages:

- Low ground clearance.
- A low-wing configuration has unstable contribution to the directional stability. Hence a larger vertical tail area is needed.



Advantages:

- Lower drag.
- Advantages of ground clearance as in the high wing configuration.
- No blockage of visibility. Hence used on some military airplanes.

Disadvantages:

- Wing root structure passing through the fuselage is not possible, which leads to higher weight.

Wing Vertical Location



Advantages:

- Allows placing fuselage closer to ground, thus allowing loading and unloading without special ground handling equipment. (good for cargo handling)
- Jet engines & propeller have sufficient ground clearance without excessive landing gear length leading to lower landing gear weight.
- For low-speed airplanes, weight saving can be affected by strut braced wing.
- For short take off and landing (STOL) airplanes with high wing configuration have following specific advantages. (a) Large wing flaps can be used (b) Engines are away from the ground and hence ingestion of debris rising from unprepared runways is avoided (c) Prevents floating of wing due to ground effect which may occur for low wing configuration.

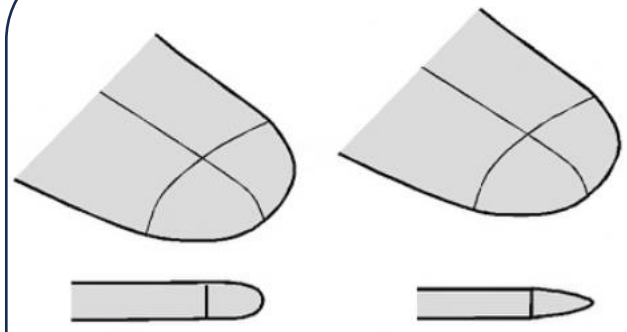
Disadvantages:

- Fuselage generally houses the landing gear in special pods leading to higher weight and drag.
- Pilot's visibility may be blocked in a turn.

Wing Tips

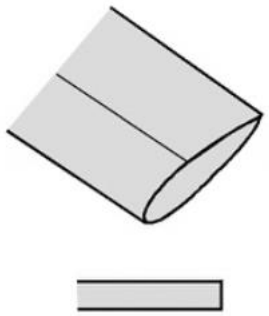
What does the wing tip shape affect?

- The aircraft wetted area, but only to a small extent.
- The tip vortices.



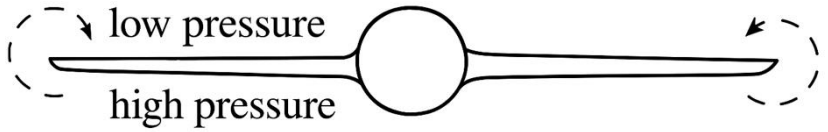
Rounded **Sharp**

- A smoothly-rounded tip easily permits the air to flow around the tip.
- A tip with sharp edge makes it more difficult, thus reducing the induced drag.



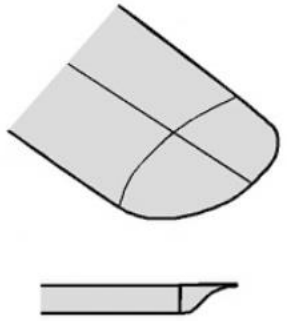
Cut-off

- Even a simple cut-off tip offers less drag than a rounded-off tip, due to the sharp edges where the upper and lower surfaces end.



This air escape near the wing tips causes induced drag.

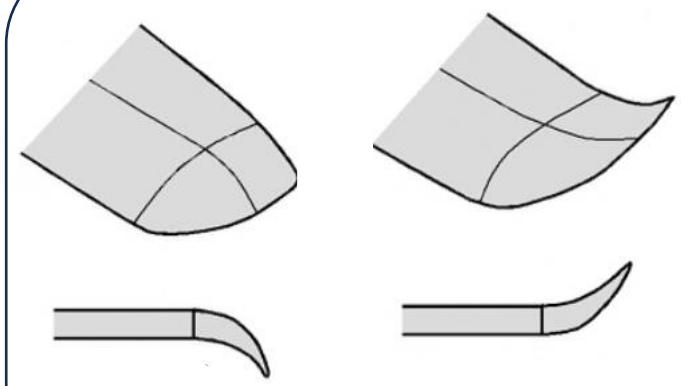
Most of the new low-drag wing tips use some form of sharp edge.



Hoerner

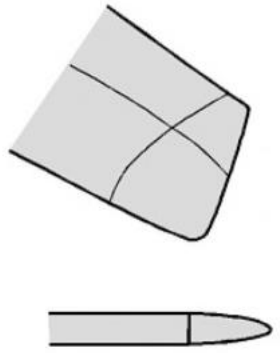
- Hoerner is the most widely used low-drag wing tip.
- The lower surface is "undercut".

Wing Tips



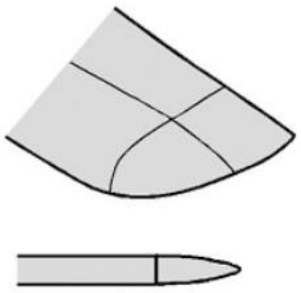
Drooped **Upswept**

- The "drooped" and "upswept" wing tips are similar to the Hoerner wing tip except that the tip is curved upwards or downwards to increase the effective span without increasing the actual span.



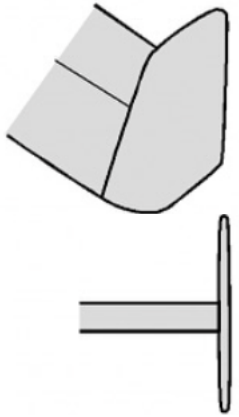
Cut-off Forward-swept

- F-15 has forward swept tips on all surfaces.
- keeps trailing edge out of tip vortex.



Aft-swept

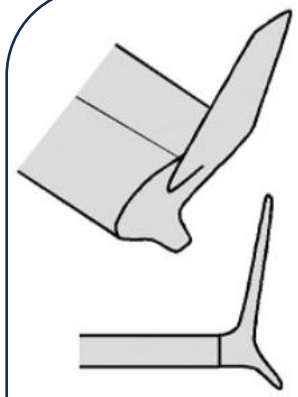
- The tip vortex tends to be located at the trailing edge: an aft-swept wig tip, with a greater trailing edge span, tends to have lower drag.
- However, it tends to increase the wing torsional loads.



End-plate

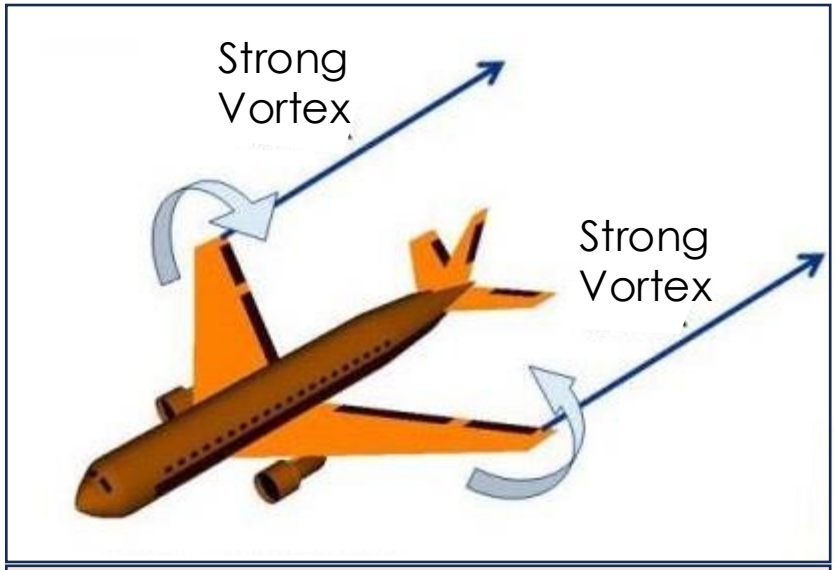
- An obvious way to prevent this would be to mount a vertical plate at the wing tip.
- The wetted area of the endplate itself creates drag.

Wing Tips

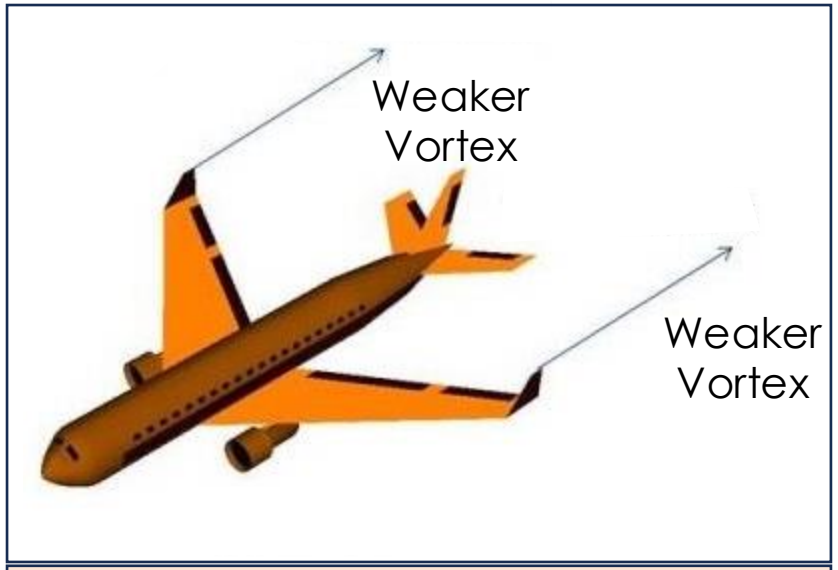


Winglet

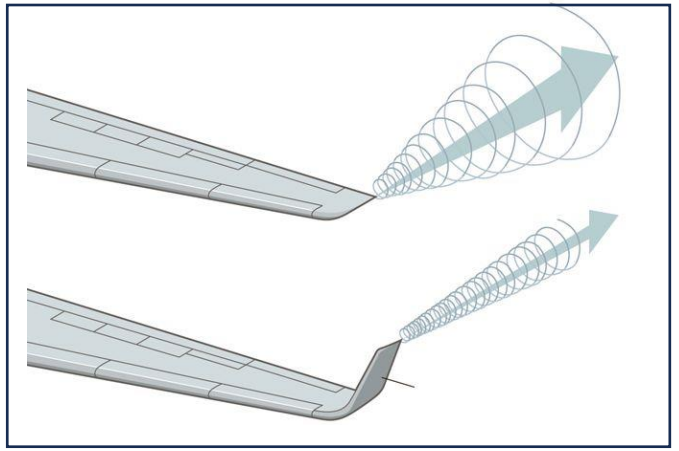
- An advanced version of the endplate can offer lower drag than an equal-area increase in wing span.
- The Winglet, gets an additional drag reduction by using the energy available in the tip vortex.
- One problem with winglets is that they add weight behind the elastic axis of the wing, which can aggravate flutter tendencies.
- Also, the twist and camber of a winglet must be optimized for one velocity. At other than design speed, the winglet will provide less benefit.



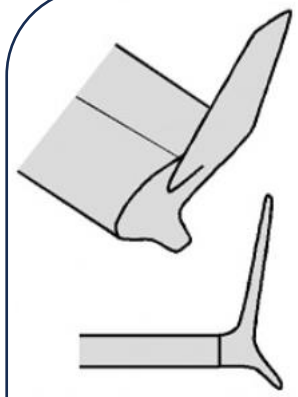
Without Winglets



With Winglets



Wing Tips



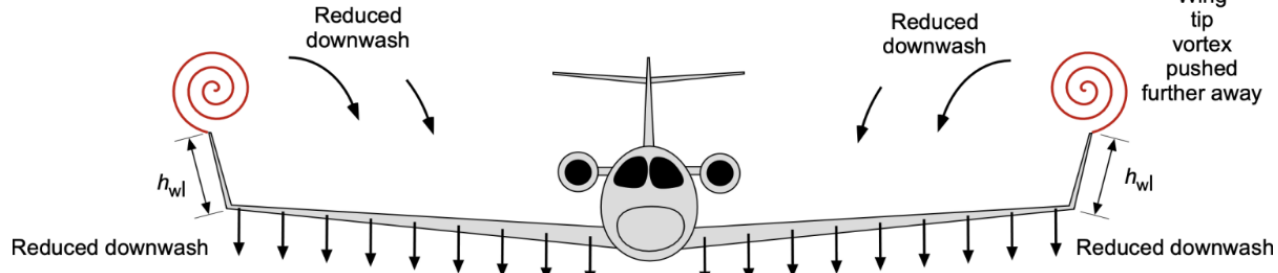
Winglet

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Without Winglets



With Winglets



Winglets provide the greatest benefit when the wing tip vortex is strong, so a low-aspect-ratio wing will see more advantage from the use of winglets than an already-efficient high-aspect-ratio wing.

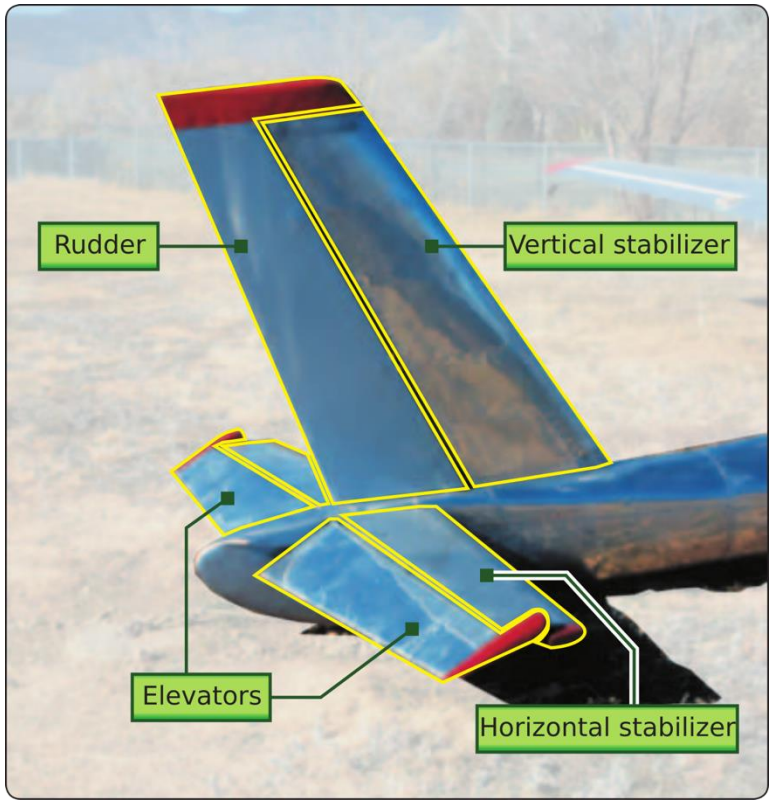
Tail Arrangements

Tail Functions

- "Tails are little wings." Much of the previous discussion concerning wings can also be applied to tail surface.

What is the difference between tail and wing?

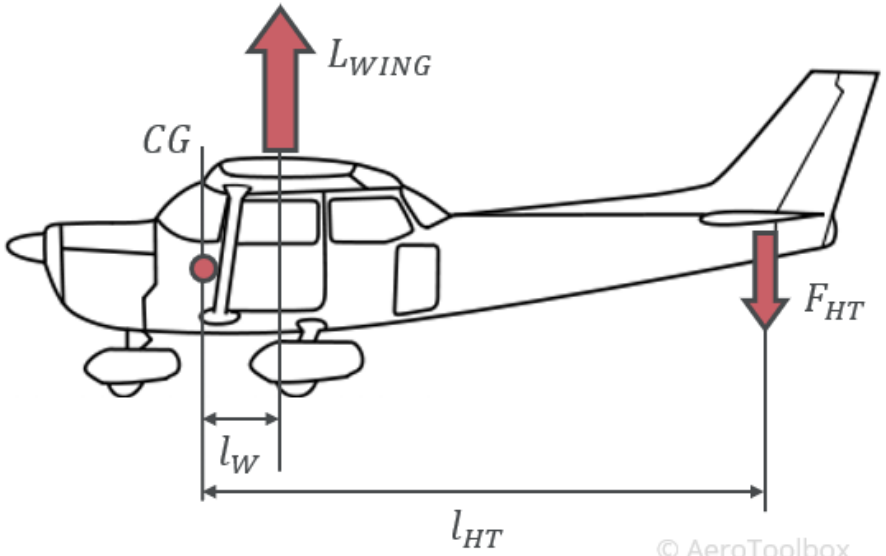
- While the wing is designed to routinely carry a substantial amount of lift, a tail is designed to operate normally at only a fraction of its lift potential.
- Any time in flight that a tail comes close to its maximum lift potential, and hence its stall angle, something is very wrong!



- Tails provide for:
  - Trim
  - Stability
  - Control.

Tail Geometry and Arrangements

Tail Functions



- Tails provide for:
  - Trim
  - Stability
  - Control.

- Trim
- For the horizontal tail, trim primarily refers to the balancing of the moment created by the wing. An aft horizontal tail typically has a negative incidence angle of about 2° – 3° to balance the wing pitching moment.
- For vertical tail, most aircraft are left-right symmetric, and so unbalanced aerodynamic yawing moments requiring trim are not created during normal flight.
  - The vertical tail of a multi-engined aircraft must be capable of providing a sufficient trim force in the event of an engine failure.

Tail Geometry and Arrangements

Tail Functions

- Tails provide for:
  - Trim
  - Stability
  - Control.

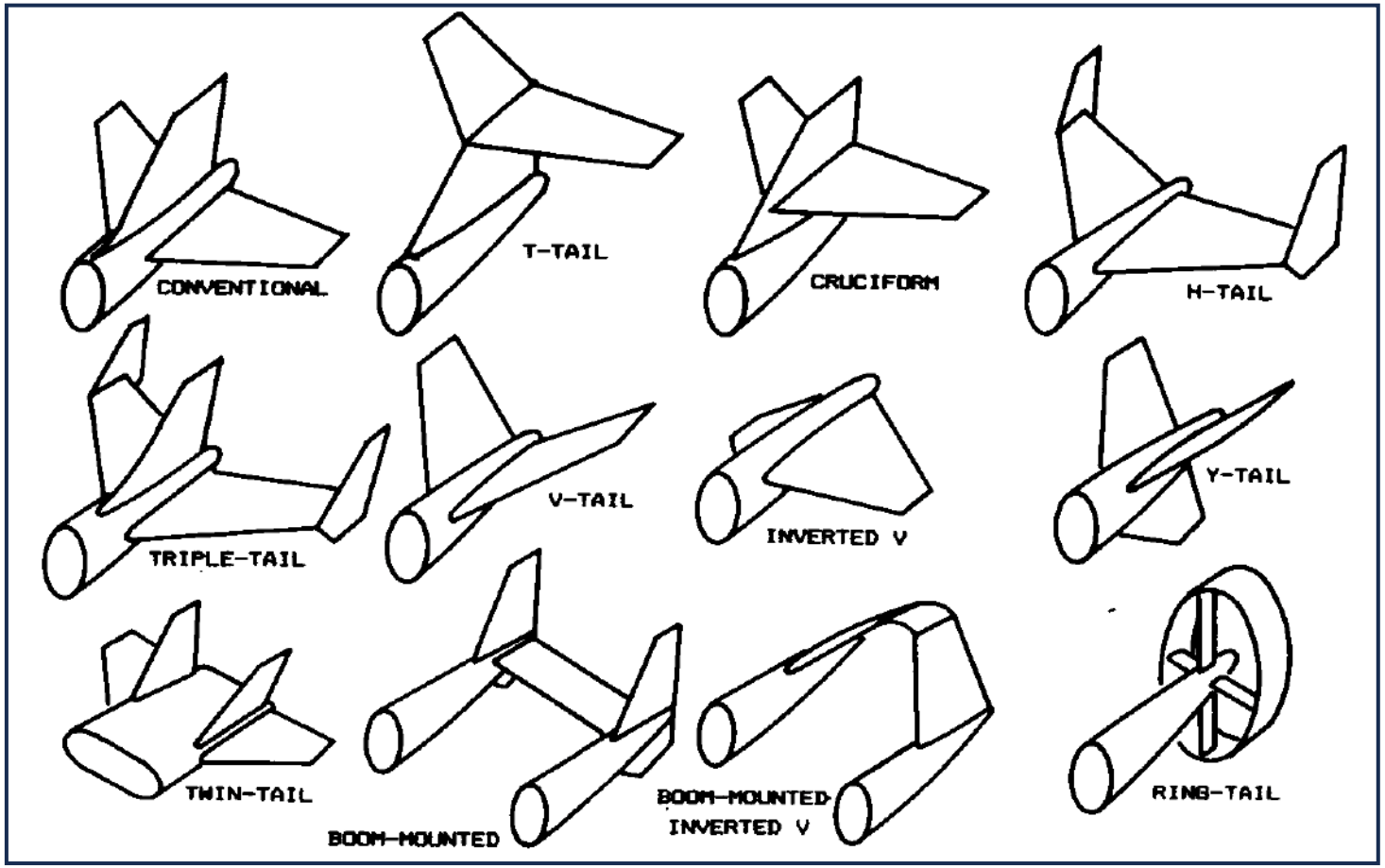
- Stability
- The tails are also a key element of stability, acting much like the fins on an arrow. While it is possible to design a stable aircraft without tails, such a design is usually penalized in some other area, such as a compromised airfoil shape, excessive wing area or sweep, or narrow center-of-gravity range.

- Control
- The other major function of the tail is control. The tail must be sized to provide adequate control power at all critical conditions. These critical conditions for the horizontal tail or canard typically include nosewheel liftoff, low-speed flight with flaps down, and transonic maneuvering. For the vertical tail, critical conditions typically include engine-out flight at low speeds, maximum roll rate, and spin recovery.

The control power depends upon the size and type of the movable surface as well as the overall size of the tail itself.

Tail Arrangements

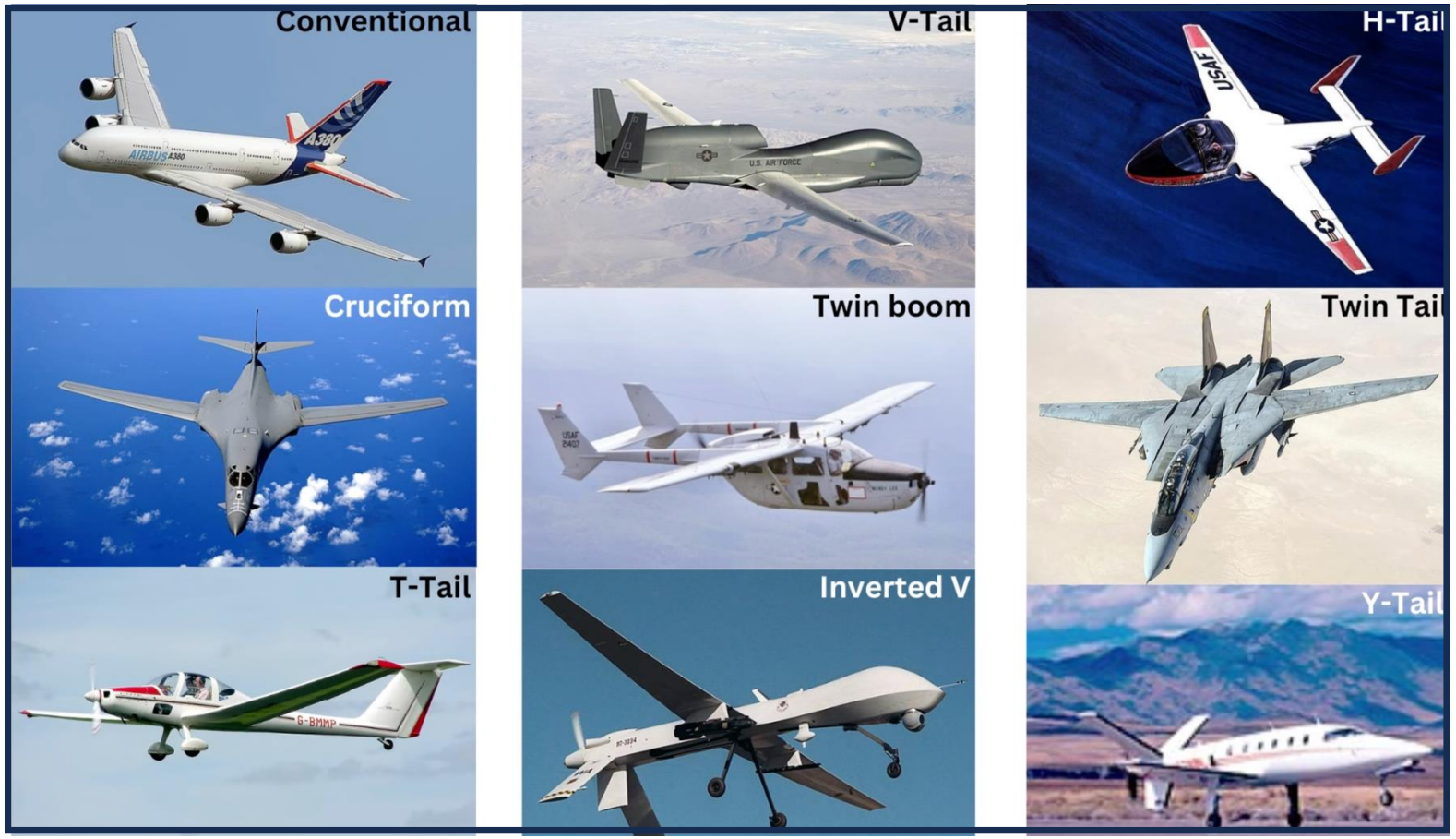
Some of the possible variations in aft-tail arrangement.



Tail Geometry and Arrangements

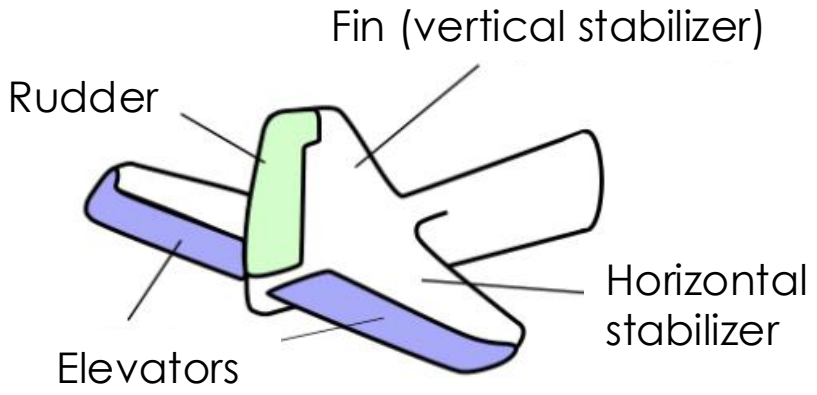
Tail Arrangements

Some of the possible variations in aft-tail arrangement.



Tail Geometry and Arrangements

Conventional Tail

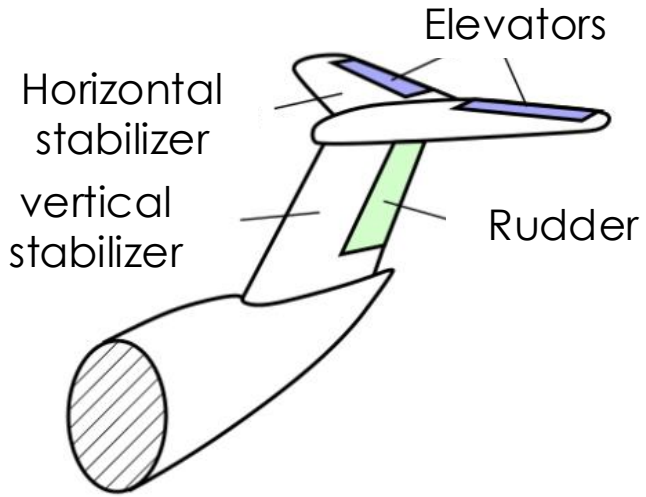


- This has become "conventional" for the simple reason that it works.
- For most aircraft designs, the conventional tail will usually provide adequate stability and control at the lightest weight.
- Probably 70% or more of the aircraft in service have such a tail arrangement.



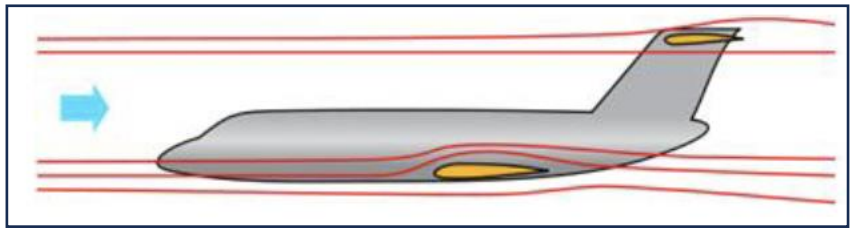
Tail Geometry and Arrangements

T-Tail





- A T-tail is inherently heavier than a conventional tail because the vertical tail must be strengthened to support the horizontal tail.

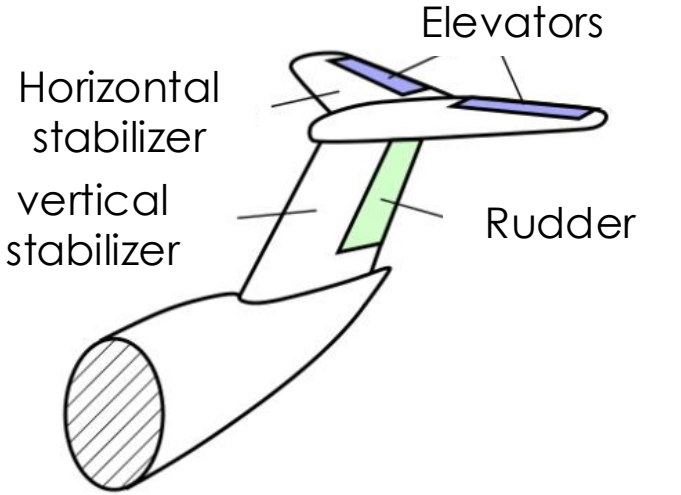


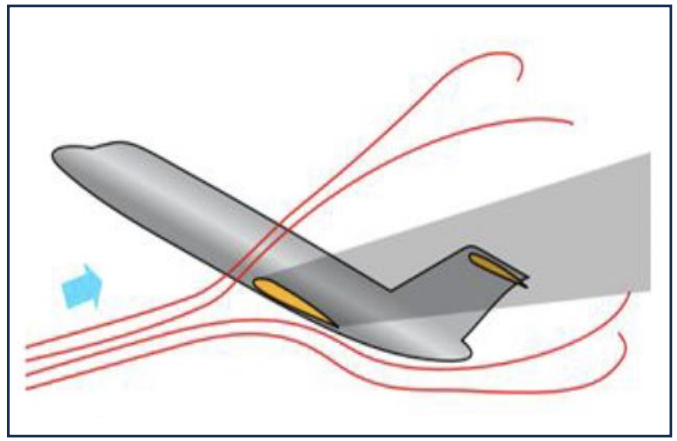
- Due to end-plate effect, the T-tail allows a smaller vertical tail. The T-tail lifts the horizontal tail clear of the wing wake and propwash, which makes it more efficient and hence allows reducing its size.
- This also reduces buffet on the horizontal tail, which reduces fatigue for both the structure and the pilot.

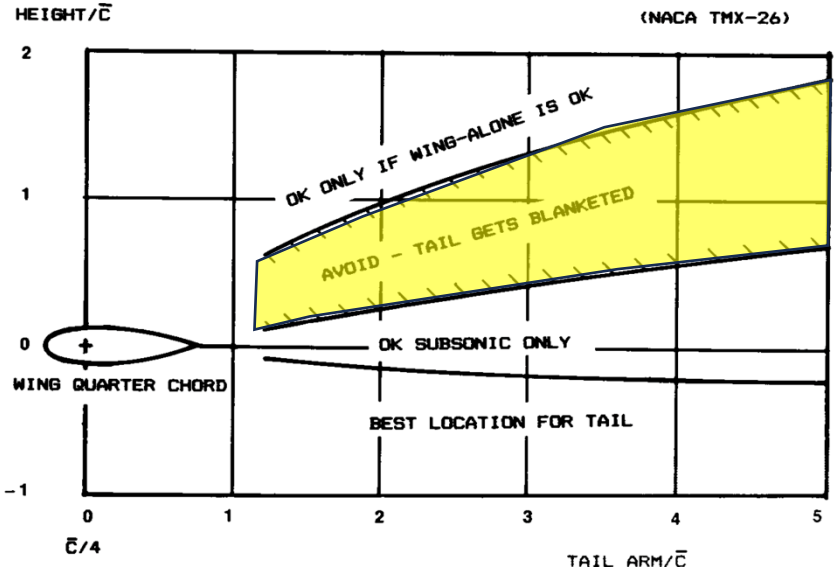
- In jet transport aircraft, the T-tail allows the use of engines mounted in pods on the aft fuselage. This increases the wing lift and decreases the wing drag.

Tail Geometry and Arrangements

T-Tail







- A deep stall is a dangerous type of stall that affects those with a T-tail configuration.
- The turbulent wake of a stalled main wing "blankets" the horizontal stabilizer, rendering the elevators ineffective and preventing the aircraft from recovering from the stall.

- The boundaries of the acceptable locations for horizontal tail to avoid this problem is given in the figure. Low tails are best for stall recovery.
- A tail approximately in line with the wing is acceptable for a subsonic aircraft, but may cause problems at supersonic speeds due to the wake of the wing.

Tail Geometry and Arrangements

Cruciform Tail



- The cruciform tail, a compromise between the conventional and T-tail arrangements, lifts the horizontal tail to avoid proximity to a jet exhaust, or to expose the lower part of the rudder to undisturbed air during high-angle of attack conditions and spins.
- It has less of a weight penalty according to T-tail.
- However, it will not provide a tail-area reduction due to endplate effect.



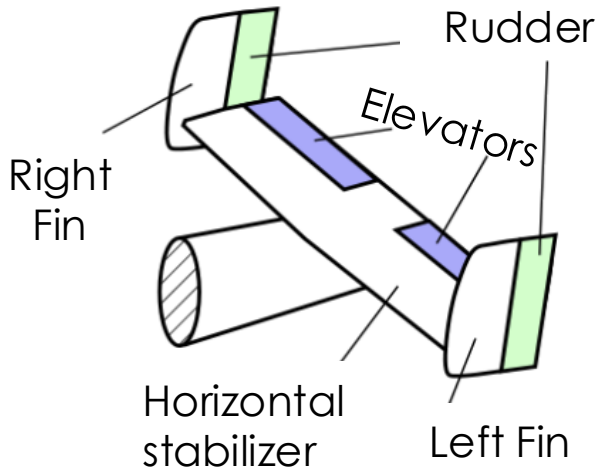
Tail Geometry and Arrangements

Cruciform Tail



- The cruciform tail, a compromise between the conventional and T-tail arrangements, lifts the horizontal tail to avoid proximity to a jet exhaust, or to expose the lower part of the rudder to undisturbed air during high-angle of attack conditions and spins.
- It has less of a weight penalty according to T-tail.
- However, it will not provide a tail-area reduction due to endplate effect.

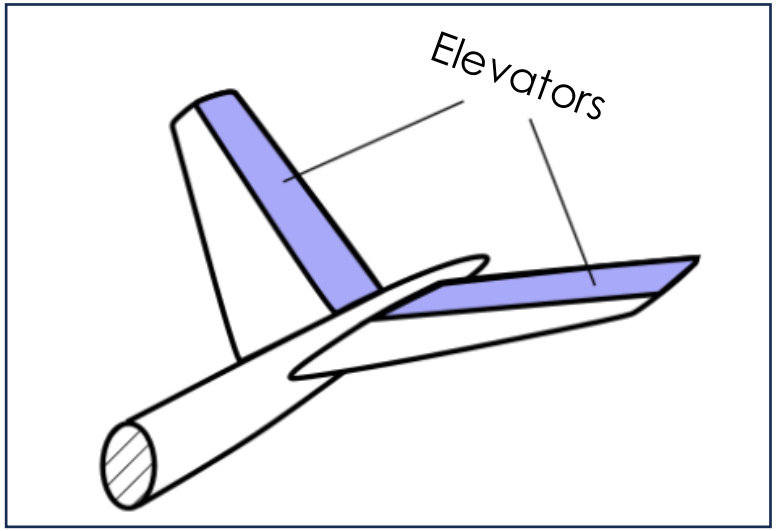
H-Tail



- The H-tail is used primarily to position the vertical tails in undisturbed air during high angle-of-attack conditions or to position the rudders in the propwash on a multiengine aircraft to enhance engine-out control.
- H-tail is heavier than the conventional tail, but its endplate effect allows a smaller horizontal tail.
- H-tails and and the related Triple-tails have also been used to lower the tail height to allow an aircraft to fit into existing hangars.

Tail Geometry and Arrangements

V-Tail



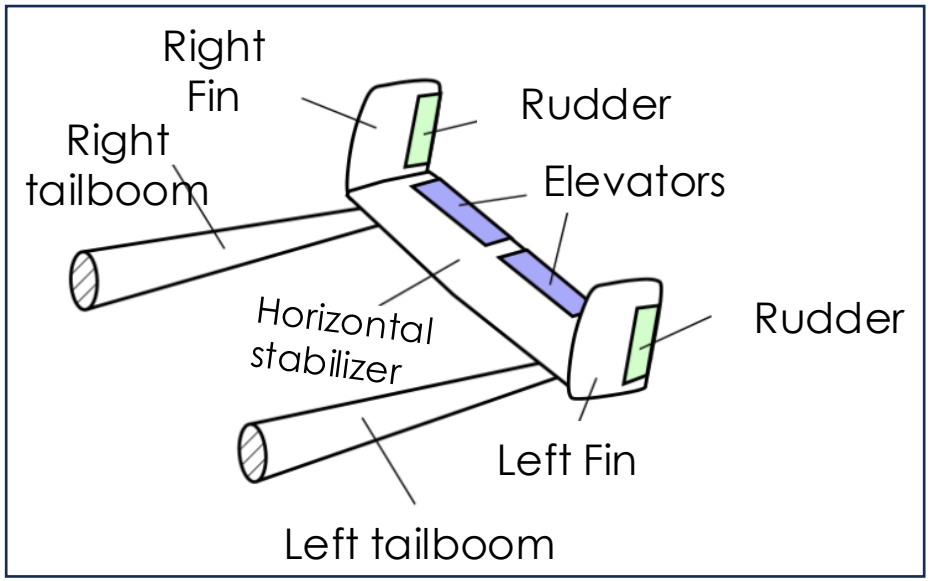


- The V-tail is intended to reduce wetted area.
- The horizontal and vertical tail forces are the result of horizontal and vertical projections of the force exerted upon the "V" surfaces.
- The tail dihedral angle would be found as the arctangent of the ratio of required vertical and horizontal areas.
- V-tails offer reduced interference drag but at some penalty in control-actuation complexity.
- The rudder and elevator control inputs must be blended in a "mixer" to provide the proper movement of the V-tail "ruddervators".

- When the right rudder pedal of a V-tail aircraft is pressed, the right ruddervator deflects downward, and the left ruddervator deflects upward. The combined forces push the tail to the left, so the nose goes to the right as desired. However, the ruddervators also produce a rolling moment toward the left—in opposition to the desired direction of turn an action called "adverse roll-yaw coupling".

Tail Geometry and Arrangements

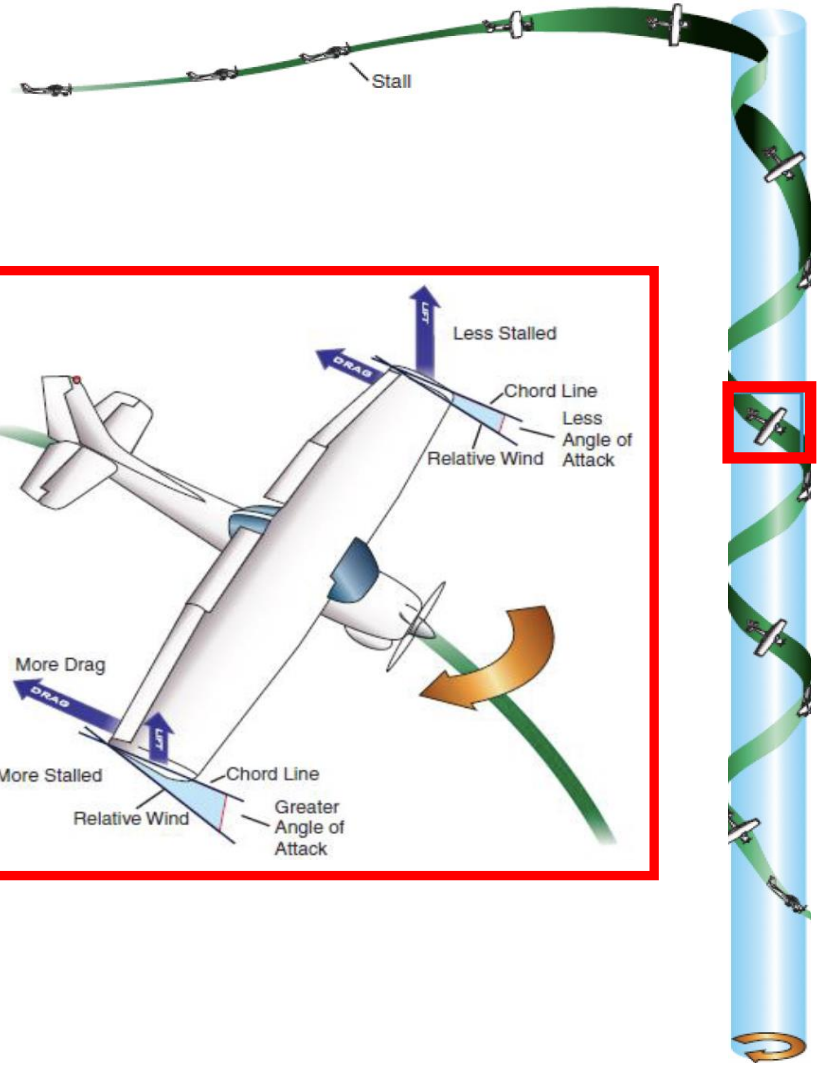
Boom Mounted

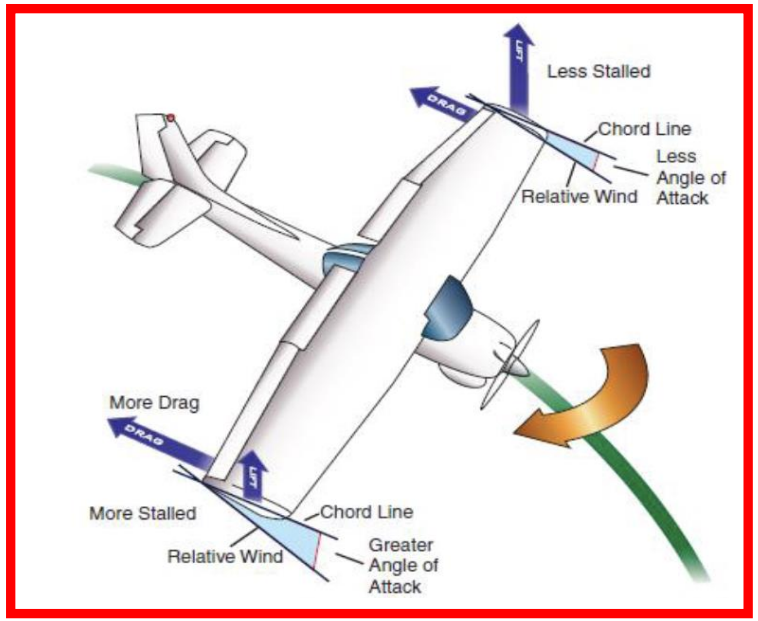


- Boom-mounted tails have been used to allow pusher propeller or allow location of a heavy jet engine near the center of gravity.
- Tailbooms are typically heavier than a conventional fuselage construction, but can desirable in some applications.

Tail Geometry and Arrangements

Tail Arrangement for Spin Recovery





INCIPIENT SPIN
- Lasts about 4 to 6 seconds in light aircraft.
- Approximately 2 turns

FULLY DEVELOPED SPIN
- Airspeed, vertical speeds, and rate of rotation are stabilized.
- Small, training aircraft lose approximately 500 feet per each 3 second turn

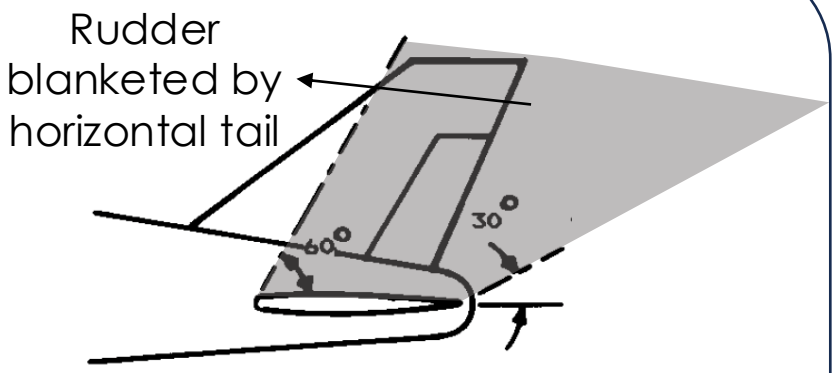
RECOVERY
- Wings regain lift.
- Training aircraft usually recover in about ¼ to ½ of a turn after antispin inputs are applied.

- The vertical tail plays a key role in spin recovery. An aircraft in a spin is essentially falling vertically and rotating about a vertical axis, with the inside wing fully stalled. The aircraft is also typically at a large sideslip angle. To recover from the spin requires that the wing will be unstalled, so the angle of attack must be reduced. However, first the rotation must be stooped and the sideslip angle reduced, or the aircraft will immediately enter another spin. This requires adequate rudder control even at the high angles of attack seen in the spin.

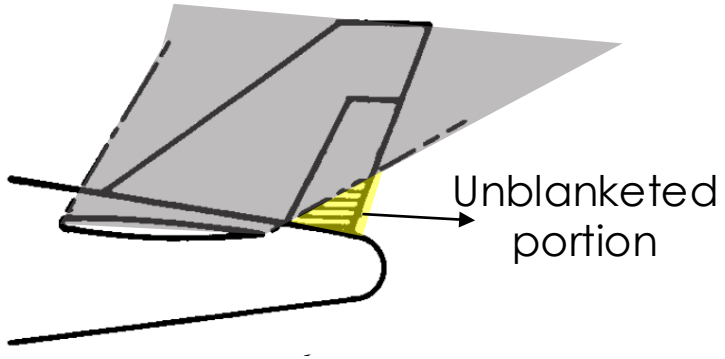
- The effect of tail arrangement upon rudder control at high angles of attack: At high angle of attack the horizontal tail is stalled, producing a turbulent wake extending upward at approximately 45°.

Tail Arrangement for Spin Recovery

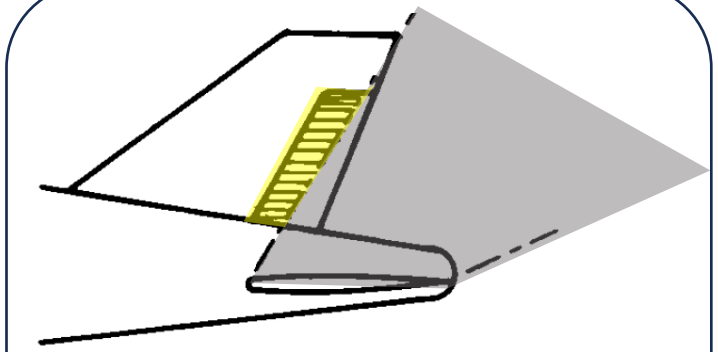
- The effect of tail arrangement upon rudder control at high angles of attack: At high angle of attack the horizontal tail is stalled, producing a turbulent wake extending upward at approximately 45°.



The rudder lies entirely within the wake of the horizontal tail.



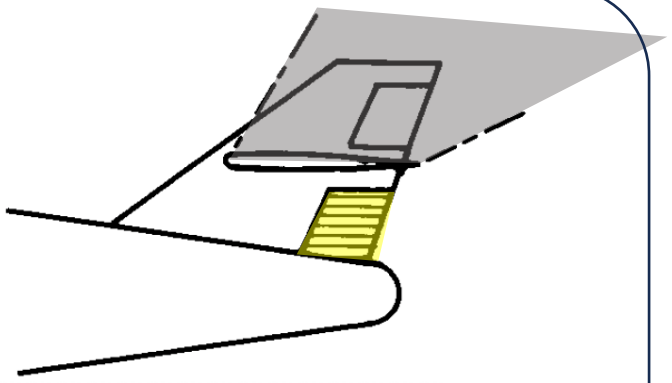
The effect of moving the horizontal tail forward with respect to the vertical tail.



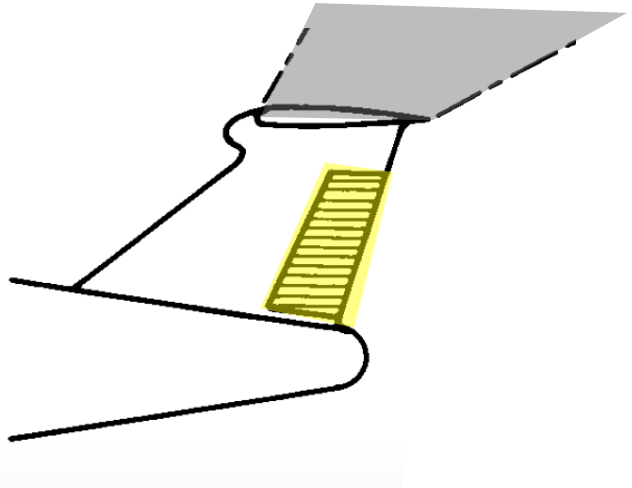
Moving of the horizontal tail aft with respect to the vertical tail.

Tail Arrangement for Spin Recovery

- The effect of tail arrangement upon rudder control at high angles of attack: At high angle of attack the horizontal tail is stalled, producing a turbulent wake extending upward at approximately 45°.



Moving the horizontal tail upward.



The T-tail arrangement completely uncovers the rudder, but can result in pitchup and loss of elevator control.

Tail Geometry and Arrangements

Tail Geometry

- The surface areas required for all types of tails are directly proportional to the aircraft's wing area.
- The tail areas can not be selected until the initial estimate of aircraft takeoff gross weight has been made.
- The initial estimation of tail area is made using the "tail volume coefficient" method.

- Other geometric parameters for the tails can be selected:
  - Tail aspect ratio and taper ratio show little variation over a wide range of aircraft types.
  - Tail aspect ratio and taper ratio.

	Horizontal tail		Vertical tail	
	A	λ	A	λ
Fighter	3 – 4	0.2 – 0.4	0.6 – 1.4	0.2 – 0.4
Sail plane	6 – 10	0.3 – 0.5	1.5 – 2.0	0.4 – 0.6
Others	3 – 5	0.3 – 0.6	1.3 – 2.0	0.3 – 0.6
T-tail	-	-	0.7 – 1.2	0.6 – 1.0

- T-tail aircraft have lower vertical-tail aspect ratios to reduce the weight impact of the horizontal tail's location on top of the vertical tail.
- Some general-aviation aircraft use untapered horizontal tails (λ = 1.0) to reduce manufacturing costs.
- Leading-edge sweep of the horizontal tail is usually set to about 5° more than the wing sweep. This tends to make the tail stall after the wing, and also provides the tail with a higher M_{cr} than the wing, which avoids loss of elevator effectiveness due to shock formation.

Tail Geometry and Arrangements

Tail Geometry

- The exact planform of the tail surfaces is actually not very critical in the early stages of the design process. The tail geometries are revised during later analytical and wind-tunnel studies. For conceptual design, it is usually acceptable simply to draw tail surfaces that "look right," based upon prior experience and similar designs.
- Tail thickness ratio is usually similar to the wing thickness ratio, as determined by the historical guidelines provided in the wing-geometry section. For a high-speed aircraft, the horizontal tail is frequently about 10% thinner than the wing to ensure that the tail has a higher M_{cr}.